NSE SIGNAL

Stock Selection & Calculation Reference

How every indicator, signal, and score is computed

Section	Topic	Inputs	Output
1	EMA Calculations	OHLC price	EMA10, EMA50H/L
2	ATR — Volatility	High, Low, Close	ATR in Rs.
3	Gap Analysis	EMA10, EMA50H, ATR	Gap in ATR units
4	RSI	Close prices	0–100 oscillator
5	MACD	Close prices	Momentum histogram
6	Volume Ratio	Volume	Ratio vs 20d avg
7	Signal Strength	All above	Rating 1–5
8	ML Model	40+ features	Predicted 5d return
9	Entry Confirmation	4 filters	GO / WAIT / AVOID
10	Fundamental Screen	screener.in data	Fund score 0–50

1. Exponential Moving Averages (EMA)

EMA gives more weight to recent prices than older ones. All EMA calculations use the standard exponential smoothing formula with the span parameter controlling how quickly weights decay.

1.1 Formula

```
EMA(today) = Close(today) x K  +  EMA(yesterday) x (1 - K)
K = 2 / (Period + 1)    [ smoothing factor ]
```

Example for EMA10: K = 2/(10+1) = 0.1818

Today gets 18.18% weight. Yesterday's EMA carries 81.82%.

1.2 EMAs Calculated

Name	Period	Applied to	Purpose
EMA_10	10 days	Close	Short-term trend — entry trigger
EMA_20	20 days	Close	Medium-term confirmation
EMA_50	50 days	Close	Medium trend baseline
EMA_200	200 days	Close	Long-term trend direction
EMA50_HIGH	50 days	High	Upper band — sustain target
EMA50_LOW	50 days	Low	Lower band — minimum support

1.3 EMA50 Band — The Key Zone

EMA50_HIGH (50-day EMA of the daily High) and EMA50_LOW (50-day EMA of the daily Low) form a dynamic band that acts as the primary support and resistance zone for this system.

- Price above EMA50_HIGH = strong bull zone
- Price between EMA50_HIGH and EMA50_LOW = in band, watch closely
- Price below EMA50_LOW = bear zone, avoid entry

2. ATR — Average True Range

ATR measures daily volatility in rupees. It tells you how much the stock typically moves in a single trading day. Used to normalise the Gap signal so it is comparable across stocks of different prices.

2.1 Formula

```
True Range = max(  High - Low,
                   |High - PrevClose|,
                   |Low  - PrevClose|  )
ATR(14) = 14-day rolling mean of True Range
```

2.2 Interpretation

ATR Level	Meaning	Impact on signals
Low ATR (< avg)	Calm, low volatility stock	Smaller gap still reliable
Avg ATR	Normal volatility	Standard thresholds apply
High ATR (> avg)	Volatile stock	Needs larger gap to confirm

3. Gap Analysis — The Core Signal

The Gap measures how far EMA10 is above or below EMA50_HIGH, expressed in ATR units. This is the single most important indicator in this system. A wider gap means more cushion — the price has more room before it risks falling below the key support level.

3.1 Three Gap Calculations

```
Gap (Rs.)    =  EMA10 - EMA50_HIGH
Gap (ATR)    =  Gap (Rs.)  /  ATR(14)    <-- main metric
Gap (%)      =  Gap (Rs.)  /  EMA50_HIGH  x  100
```

Gap (ATR) is the metric used for all signal scoring — it adjusts for the stock's volatility.

A Rs.10 gap means very different things for a Rs.150 stock vs a Rs.1000 stock.

But +1.0x ATR means the same cushion regardless of stock price.

3.2 Gap Signal Thresholds

Gap (ATR)	Zone	Signal	Meaning
> +2.0x	Strong bull	STRONG GO	EMA10 far above 50H — best cushion
+1.0 to +2.0x	Good cushion	GO	Solid buffer above level
+0.5 to +1.0x	Thin cushion	WAIT	Some buffer but weak
0 to +0.5x	Barely above	AVOID	Any dip breaks below 50H
< 0	Below 50H	AVOID	Wait for recovery

3.3 Why Gap Determines Streak Duration

Historical analysis shows: stocks that cross above EMA50_HIGH with a gap of +1 ATR or more tend to sustain above the level for significantly longer. The scatter plot in Cell 15 (gap at streak start vs streak duration) shows approximately +11 days of additional duration per 1 ATR unit of initial gap.

4. RSI — Relative Strength Index

RSI measures the speed and magnitude of recent price changes. It oscillates between 0 and 100. The key use here is identifying the RSI level at the moment EMA10 crosses above EMA50_HIGH — crosses from lower RSI tend to sustain longer than crosses from overbought RSI.

4.1 Formula

```
RSI = 100 - ( 100 / (1 + RS) )
RS  = Average Gain(14) / Average Loss(14)
```

Calculated over a 14-day rolling window by default. Also calculated over 9 days as RSI_9.

4.2 RSI Signal Zones

RSI Range	Zone	Signal	Why
< 30	Oversold	Strong GO	Price fell too far, rebound likely
30 – 45	Recovery	GO	Building momentum from low
45 – 60	Ideal zone	BEST ENTRY	Not overbought, trending up
60 – 70	Approaching OB	WAIT	Late to the move
> 70	Overbought	AVOID	High reversal risk at entry

5. MACD — Moving Average Convergence Divergence

MACD tracks the relationship between two EMAs of the close price. The histogram shows the difference between MACD and its signal line — positive and growing means momentum is accelerating upward.

5.1 Formula

```
MACD Line   = EMA(12) - EMA(26)
Signal Line = EMA(9) of MACD Line
MACD Hist   = MACD Line - Signal Line
```

5.2 MACD Signal Rules

Condition	Signal	Meaning
Hist > 0 AND MACD > 0	GO	Both momentum and trend positive
Hist > 0 only	WAIT	Short-term momentum positive
Hist < 0	AVOID	Momentum weakening or negative

6. Volume Ratio

Volume confirmation tells you whether the breakout above EMA50_HIGH has institutional backing. High volume at a cross = real demand, not just noise.

6.1 Formula

```
Vol Ratio = Volume(today) / SMA(Volume, 20)
```

SMA(Volume, 20) = simple 20-day average of daily volume.

Vol Ratio = 1.0 means today's volume equals the 20-day average.

6.2 Volume Signal Thresholds

Vol Ratio	Signal	Meaning
>= 1.5x	STRONG GO	50%+ above average — strong institutional buying
1.2x to 1.5x	GO	Above average — good participation
0.8x to 1.2x	WAIT	Average volume — move may not sustain
< 0.8x	AVOID	Below average — weak breakout, likely fake

7. Signal Strength Rating 1 – 5

Every time EMA10 crosses above EMA50_HIGH, a strength score is computed from 5 factors. Each factor contributes partial points for finer granularity. The total maps to a rating of 1 (very weak) to 5 (exceptional).

7.1 The 5 Factors

Factor	Condition	Full (1.0 pt)	Half (0.5 pt)
Gap (ATR)	EMA10 cushion above 50H	gap >= 1.5x	gap >= 0.5x
RSI zone	Entry RSI position	30 < RSI <= 60	60 < RSI <= 72
MACD	Momentum direction	Hist>0 AND MACD>0	Hist > 0 only
Volume	Breakout participation	vol >= 1.5x avg	vol >= 1.0x avg
Slope	10-day price slope	slope > 0 (rising)	N/A

7.2 Strength Rating Map

```
Total score (0 - 5.0)  →  Rating
  >= 4.5  →  5  (Exceptional — all 5 factors aligned)
  >= 3.5  →  4  (Strong — 4 factors confirmed)
  >= 2.5  →  3  (Moderate — 3 factors confirmed)
  >= 1.5  →  2  (Weak — only 2 factors)
  <  1.5  →  1  (Very weak — avoid entry)
```

7.3 Visual on Chart

Rating	Marker	Colour	Action
5	Large ▲	Yellow #ffea00	Strong entry — enter
4	Large ▲	Lime green	Good entry — enter
3	Medium ▲	Sky blue	Acceptable — enter with caution
2	Small ▲ (hollow)	Grey	Weak — filtered, skip
1	Tiny ▲ (hollow)	Dark grey	Fake — ignore

The strength number is printed inside each triangle on the PDF chart.

Exit triangles (▼) use the same red colour regardless of exit strength.

8. Machine Learning Model

A regression model is trained to predict the 5-day forward percentage change in the stock's closing price. The model is trained on historical data using a strict chronological train/test split to prevent data leakage.

8.1 Target Variable

```
Target = ((Close(t+5) - Close(t)) / Close(t)) * 100
```

5-day forward return in % — what the stock will do over the next week.

8.2 Train / Test Split (No Data Leakage)

```
Train set = First 80% of data  (chronological order)
Test set  = Last  20% of data  (model NEVER saw this data)
```

The model is first evaluated on the held-out test set to measure real accuracy.

Then retrained on ALL data before making future predictions.

8.3 Models Compared

Model	Strength	Best for
RandomForest	Robust to outliers, handles nonlinearity	Most NSE stocks
LightGBM	Fast, handles many features well	Large datasets
XGBoost	Strong ensemble, regularised	Clean data
GradBoost	Good general performer	Medium datasets
HuberRidge	Robust to extreme returns	Simple trends

Best model selected by highest Direction Accuracy on the held-out test set, with MAE as tiebreaker.

8.4 Feature Set (40+ indicators)

Category	Features
EMA Ratios	EMA10_RATIO, EMA20_RATIO, EMA50_RATIO, SMA200_RATIO, EMA50H_RATIO, EMA50L_RATIO
Gap Features	GAP_ATR, GAP_PCT, CLOSE_GAP_ATR
Trend Direction	TREND_5D, TREND_10D, TREND_20D, ABOVE_EMA50H, ABOVE_EMA50L
Momentum	RSI_14, RSI_9, MACD_HIST, MACD_CROSS, STOCH_K, STOCH_D
Volatility	ATR_PCT, VOL_5D, VOL_20D, BB_WIDTH, BB_PCT_B
Volume	VOL_RATIO, VWAP_DEV
Lag Returns	RET_1D, RET_2D, RET_3D, RET_5D, RET_10D, RET_20D

Category	Features
Slope	SLOPE_10D, SLOPE_20D
Calendar	DOW (day of week), MONTH

8.5 How to Read Accuracy Metrics

Metric	What it measures	Good value	HPCL example
MAE	Avg prediction error (%change)	< 2.0%	2.96%
RMSE	Penalises large errors more	< 3.0%	3.77%
R2	Explained variance (0=mean, 1=perfect)	> 0.3	~0 (normal)
DirAcc%	Correct UP/DOWN direction	> 60%	61.1%
MAPE%	% error — IGNORE for % targets	N/A	Ignore

R2 near zero is NORMAL for stock % change prediction — the target is inherently noisy. Direction Accuracy > 60% on unseen data provides a real edge.

MAPE is meaningless when the true value is near zero (e.g. +0.1% return gives infinite % error).

9. Entry Confirmation — GO / WAIT / AVOID

Four filters are checked at every EMA10 cross above EMA50_HIGH. Each filter scores 0, 1, or 2. Total score (0–8) determines the overall signal.

9.1 Four Filters

Filter	Metric	GO (2 pts)	WAIT (1 pt)	AVOID (0)
Gap	EMA10 vs 50H (ATR)	>= +1.0x	+0.3 to +1.0x	< +0.3x
RSI	RSI(14) at cross	<= 60	60 – 70	> 70
MACD	Hist + MACD line	Both > 0	Hist > 0	Hist <= 0
Volume	Vol / 20d avg	>= 1.2x	0.8 – 1.2x	< 0.8x

9.2 Overall Signal

```
Total = Gap score + RSI score + MACD score + Volume score   (0 – 8)
```

STRONG BUY	Score 7–8 | All filters green | Enter immediately
BUY	Score 5–6 | Most filters passed | Enter with normal position size
WATCH	Score 3–4 | Mixed signals | Set alert, wait for improvement
AVOID	Score 0–2 | Too many filters failed | Skip this trade

9.3 Exit Rules

Trigger	Action	Priority
Close below EMA50_LOW	Exit immediately next open	Mandatory
Gap(ATR) drops below 0	Exit — EMA10 lost 50H support	Mandatory
RSI > 80 + gap narrowing	Take 50% profit	High
MACD Hist turns negative	Tighten stop to EMA10	High
Gap narrows below +0.3x	Prepare to exit	Medium

Stop Loss = EMA50_LOW (dynamic — recalculates each day as price moves).

10. Fundamental Screening

Before any technical analysis, stocks are filtered using fundamental criteria from screener.in. Revenue/sales growth carries the most weight because it is the leading indicator of a company's trajectory.

10.1 Screener.in Query

```
Market Capitalization > 2000 AND
Market Capitalization < 200000 AND
Sales growth > 15 AND
Sales growth 3Years > 12 AND
Profit growth > 10 AND
Return on equity > 12 AND
Debt to equity < 1 AND
Current ratio > 1 AND
Price to Earning < 40 AND
EPS last year > 0 AND
Return on capital employed > 12
```

Universe: Nifty 100 + Nifty Midcap 150 (MCap Rs.2,000 – 2,00,000 Cr)

10.2 Fundamental Scoring (0–50 points)

Factor	Max pts	Full pts	Half pts	Why
Sales Growth YoY	18	>=30%	15–29%	Primary filter — revenue expansion
Sales Growth 3-Year	4	>=20%	12–19%	Consistent, not one-quarter blip
Acceleration bonus	+1	Recent > 3Yr x1.2	—	Accelerating growth rewarded
Profit Growth	10	>=20%	10–19%	Profits following revenue
Operating Profit Growth	5	>=25%	10–24%	Operating leverage confirmed
ROE	6	>=25%	15–24%	Quality of growth
ROCE	4	>=20%	12–19%	Capital efficiency
Debt/Equity	3	<=0.2	0.2–0.5	Low debt = low risk
TOTAL	50	—	—	—

10.3 Combined Score — Total 0–100

```
Fundamental Score (0-50) + Technical Score (0-50) = Total (0-100)
```

Total Score	Rating	Action	Description
80 – 100	STRONG BUY	Enter now	Strong on both fund + tech
65 – 79	BUY	Enter	Solid on most criteria
50 – 64	WATCH	Set alert	Wait for technical improvement
< 50	AVOID	Skip	Too many weaknesses